

4.5 Energy, work and power

Mechanics · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Work by a force is component along displacement times distance. Net work changes kinetic energy; gravitational potential and elastic stores can simplify a mechanics problem when losses are absent. Power is work per second or instantaneous Fv when force acts along velocity.

Worked example: follow the reasoning

A 500 N force along motion at 4 m s^{-1} delivers instantaneous power 2.0 kW.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A 2 kg object falls 5 m from rest without resistance, $g=10$. Determine speed using energy conservation. [3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

Loss in GPE = $mg\Delta h = 100 \text{ J}$ [1]; $\frac{1}{2}mv^2 = 100$ [1]; $v = 10 \text{ m s}^{-1}$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Mathematics 9709, 2026–2027. <https://www.cambridgeinternational.org/Images/697427-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.